MT-AIML, Honors AIML, Sem 5, Module 4 — Collocation Mining & Association Scoring, Problem 17

Evaluation Report

This report compares association scores as collocation rankers on transcribed municipal meetings. The numbers are the pipeline's report files for this run. Precision@20 is a hit rate against a curated gold list that is not exhaustive, so it is a lower bound on precision against every true collocation.

Corpus

Source: MeetingBank, retrieved 2026-10-05, license CC BY-NC-SA 4.0 as declared on the Hugging Face dataset card (https://huggingface.co/datasets/huuuyeah/meetingbank). The assignment brief called MeetingBank CC BY 4.0. The card on the snapshot that was downloaded says CC BY-NC-SA 4.0. The non-commercial and share-alike terms are the ones that apply. AMI was inspected first and not used: the public parquet shards embed audio. VoxPopuli English shards are larger for the same reason. Only MeetingBank transcript text was kept.

The analysis set is 28 agenda-item transcripts selected deterministically from MeetingBank test.json (whitespace length between 1400 and 3200, round-robin by city, stop at 65000 whitespace tokens). Raw whitespace tokens: 66924. After preprocessing: 28 documents, 5318 sentences, 67046 tokens, 4437 lemmas. Bigram tokens N = 61728, bigram types = 27358 of which 3864 have frequency at least 3. Trigram tokens = 56858, trigram types = 45410 of which 1836 have frequency at least 3. Cities in the sample: Alameda, Boston, Denver, King County, Long Beach, and Seattle. These are segment transcripts, not full multi-hour meetings, and they contain recognition errors and thin punctuation.

ngram	category	gold phrases
2	business	12
2	meeting	38
2	project	37
2	technical	17
3	business	4
3	meeting	12
3	project	17
3	technical	7

Gold total 144: 104 bigrams and 40 trigrams. Inclusion required frequency at least 3 in the lemmatized stream and a judgment that the phrase is a fixed domain, procedural, or name unit. Grammatical bigrams and compositional time expressions were rejected. The annotator notes record the procedure and the lower-bound caveat.

Metric comparison

precision@20 = |top 20 ∩ gold| / 20. recall@20 = |top 20 ∩ gold| / |gold|. F1@20 is the harmonic mean. Bigram rows use the bigram gold list. Trigram rows use the trigram gold list.

metric	gold	P@20	R@20	F1@20	hits	mean f	top phrase
PMI	bigram	0.850	0.163	0.274	17	3.9	coffee shop
PMI^2	bigram	0.850	0.163	0.274	17	3.9	coffee shop
PMI^3	bigram	0.850	0.163	0.274	17	3.9	coffee shop
NPMI	bigram	0.950	0.183	0.306	19	7.8	boot camp
Chi-square	bigram	0.950	0.183	0.306	19	8.4	boot camp
Chi-square (Yates)	bigram	0.900	0.173	0.290	18	8.8	dry ice
Student's t	bigram	0.000	0.000	0.000	0	203.0	thank you
Log-likelihood	bigram	0.150	0.029	0.048	3	166.5	thank you
NLTK PMI	bigram	0.800	0.154	0.258	16	3.9	coffee shop
NLTK chi-square	bigram	0.950	0.183	0.306	19	8.3	boot camp

metric	gold	P@20	R@20	F1@20	hits	mean f	top phrase
NLTK t	bigram	0.000	0.000	0.000	0	203.0	thank you
NLTK G^2	bigram	0.150	0.029	0.048	3	175.2	thank you
Trigram PMI	trigram	0.800	0.400	0.533	16	3.5	vince lombardi trophy
NLTK trigram PMI	trigram	0.800	0.400	0.533	16	3.5	vince lombardi trophy
NLTK trigram chi-square	trigram	0.850	0.425	0.567	17	4.0	vince lombardi trophy
NLTK trigram t	trigram	0.000	0.000	0.000	0	46.9	be go to
NLTK trigram G^2	trigram	0.000	0.000	0.000	0	14.8	thank you mr

Best from-scratch bigram metric by F1@20: NPMI at 0.306. NPMI and chi-square lead because their top 20 is dominated by exclusive names and compounds that were gold-listed. Student's t scores 0 hits: the top 20 is grammatical and lemmatized copula bigrams. Log-likelihood sits between them, with a few real titles inside a list that is still mostly discourse.

Precision at k

metric	gold	k	hits	P	R	F1
PMI	bigram	10	10	1.000	0.096	0.175
PMI	bigram	20	17	0.850	0.163	0.274
PMI	bigram	50	34	0.680	0.327	0.442
PMI^2	bigram	10	10	1.000	0.096	0.175
PMI^2	bigram	20	17	0.850	0.163	0.274
PMI^2	bigram	50	34	0.680	0.327	0.442
PMI^3	bigram	10	10	1.000	0.096	0.175
PMI^3	bigram	20	17	0.850	0.163	0.274
PMI^3	bigram	50	34	0.680	0.327	0.442
NPMI	bigram	10	9	0.900	0.087	0.158
NPMI	bigram	20	19	0.950	0.183	0.306
NPMI	bigram	50	39	0.780	0.375	0.506
Chi-square	bigram	10	9	0.900	0.087	0.158
Chi-square	bigram	20	19	0.950	0.183	0.306
Chi-square	bigram	50	35	0.700	0.337	0.455
Chi-square (Yates)	bigram	10	9	0.900	0.087	0.158
Chi-square (Yates)	bigram	20	18	0.900	0.173	0.290
Chi-square (Yates)	bigram	50	37	0.740	0.356	0.481
Student's t	bigram	10	0	0.000	0.000	0.000
Student's t	bigram	20	0	0.000	0.000	0.000
Student's t	bigram	50	1	0.020	0.010	0.013
Log-likelihood	bigram	10	1	0.100	0.010	0.018
Log-likelihood	bigram	20	3	0.150	0.029	0.048
Log-likelihood	bigram	50	10	0.200	0.096	0.130
NLTK PMI	bigram	10	10	1.000	0.096	0.175
NLTK PMI	bigram	20	16	0.800	0.154	0.258
NLTK PMI	bigram	50	33	0.660	0.317	0.429
NLTK chi-square	bigram	10	9	0.900	0.087	0.158
NLTK chi-square	bigram	20	19	0.950	0.183	0.306
NLTK chi-square	bigram	50	36	0.720	0.346	0.468
NLTK t	bigram	10	0	0.000	0.000	0.000
NLTK t	bigram	20	0	0.000	0.000	0.000
NLTK t	bigram	50	1	0.020	0.010	0.013
NLTK G^2	bigram	10	1	0.100	0.010	0.018
NLTK G^2	bigram	20	3	0.150	0.029	0.048

metric	gold	k	hits	P	R	F1
NLTK G^2	bigram	50	10	0.200	0.096	0.130
Trigram PMI	trigram	10	10	1.000	0.250	0.400
Trigram PMI	trigram	20	16	0.800	0.400	0.533
Trigram PMI	trigram	50	23	0.460	0.575	0.511
NLTK trigram PMI	trigram	10	10	1.000	0.250	0.400
NLTK trigram PMI	trigram	20	16	0.800	0.400	0.533
NLTK trigram PMI	trigram	50	23	0.460	0.575	0.511
NLTK trigram chi-square	trigram	10	10	1.000	0.250	0.400
NLTK trigram chi-square	trigram	20	17	0.850	0.425	0.567
NLTK trigram chi-square	trigram	50	23	0.460	0.575	0.511
NLTK trigram t	trigram	10	0	0.000	0.000	0.000
NLTK trigram t	trigram	20	0	0.000	0.000	0.000
NLTK trigram t	trigram	50	3	0.060	0.075	0.067
NLTK trigram G^2	trigram	10	0	0.000	0.000	0.000
NLTK trigram G^2	trigram	20	0	0.000	0.000	0.000
NLTK trigram G^2	trigram	50	0	0.000	0.000	0.000

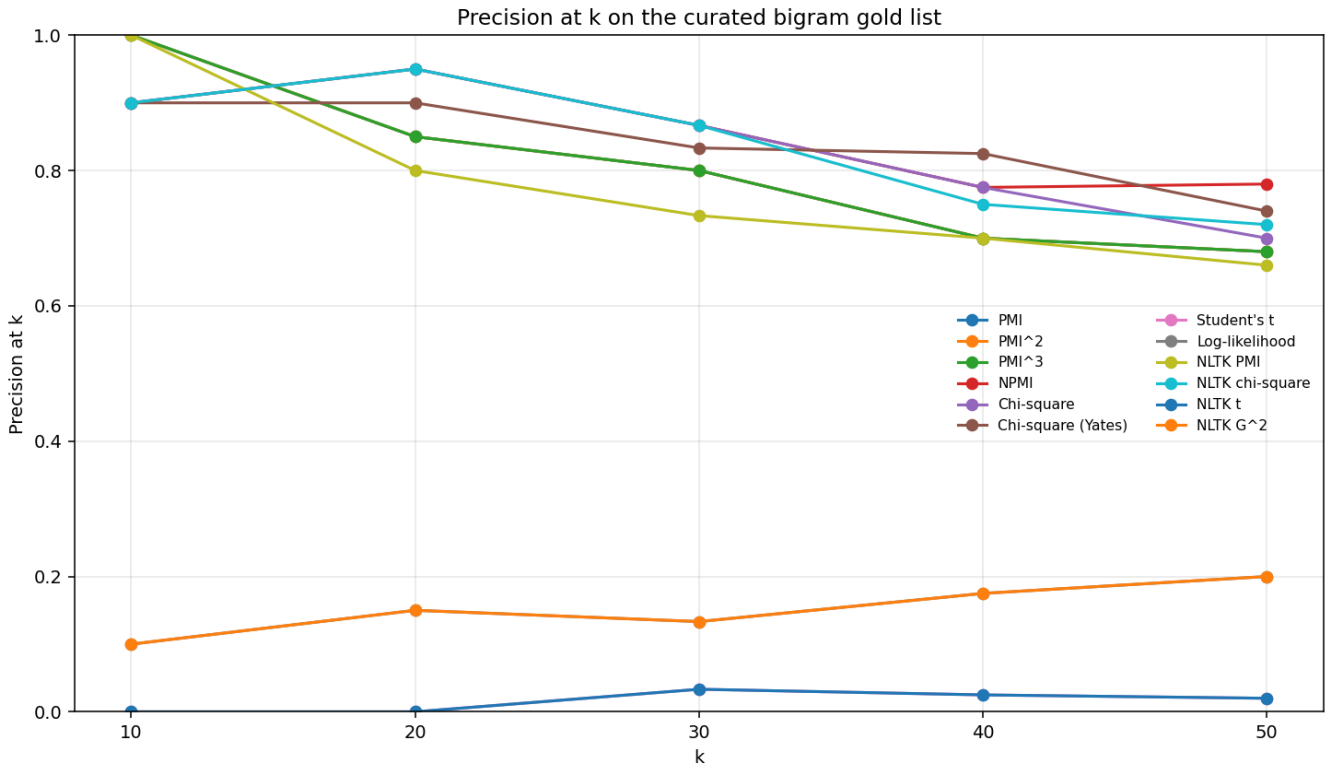


Figure. Precision at k for every bigram metric, including the NLTK variants.

Rank correlation

Spearman correlation on 3864 shared filtered bigrams. PMI versus Student's t is 0.652. PMI versus log-likelihood is 0.785. PMI versus chi-square is 0.913. PMI versus PMI squared is 0.971, below 1 because squaring folds negative PMI upward. PMI versus PMI cubed is a perfect monotone. PMI versus NLTK PMI is 0.991. Chi-square versus NLTK chi-square is 0.990. Student's t versus NLTK t is 0.990. G^2 versus NLTK's likelihood ratio is 0.989.

metric A	metric B	Spearman
PMI	PMI^2	0.9711

metric A	metric B	Spearman
PMI	PMI^3	1.0000
PMI	NPMI	0.9929
PMI	Chi-square	0.9129
PMI	Chi-square (Yates)	0.8924
PMI	Student's t	0.6524
PMI	Log-likelihood	0.7848
PMI	NLTK PMI	0.9909
PMI	NLTK chi-square	0.9001
PMI	NLTK t	0.6609
PMI	NLTK G^2	0.7717
PMI^2	PMI^3	0.9711
PMI^2	NPMI	0.9638
PMI^2	Chi-square	0.9406
PMI^2	Chi-square (Yates)	0.9242
PMI^2	Student's t	0.6137
PMI^2	Log-likelihood	0.8420
PMI^2	NLTK PMI	0.9615
PMI^2	NLTK chi-square	0.9305
PMI^2	NLTK t	0.6223
PMI^2	NLTK G^2	0.8317
PMI^3	NPMI	0.9929
PMI^3	Chi-square	0.9129
PMI^3	Chi-square (Yates)	0.8924
PMI^3	Student's t	0.6524
PMI^3	Log-likelihood	0.7848
PMI^3	NLTK PMI	0.9909
PMI^3	NLTK chi-square	0.9001
PMI^3	NLTK t	0.6609
PMI^3	NLTK G^2	0.7717
NPMI	Chi-square	0.9427
NPMI	Chi-square (Yates)	0.9263
NPMI	Student's t	0.7157
NPMI	Log-likelihood	0.8337
NPMI	NLTK PMI	0.9845
NPMI	NLTK chi-square	0.9297
NPMI	NLTK t	0.7232
NPMI	NLTK G^2	0.8199
Chi-square	Chi-square (Yates)	0.9984
Chi-square	Student's t	0.7879
Chi-square	Log-likelihood	0.9644
Chi-square	NLTK PMI	0.9065
Chi-square	NLTK chi-square	0.9898
Chi-square	NLTK t	0.7923
Chi-square	NLTK G^2	0.9521
Chi-square (Yates)	Student's t	0.8035
Chi-square (Yates)	Log-likelihood	0.9746
Chi-square (Yates)	NLTK PMI	0.8866
Chi-square (Yates)	NLTK chi-square	0.9888
Chi-square (Yates)	NLTK t	0.8069
Chi-square (Yates)	NLTK G^2	0.9626

metric A	metric B	Spearman
Student's t	Log-likelihood	0.8345
Student's t	NLTK PMI	0.6508
Student's t	NLTK chi-square	0.7744
Student's t	NLTK t	0.9900
Student's t	NLTK G^2	0.8145
Log-likelihood	NLTK PMI	0.7806
Log-likelihood	NLTK chi-square	0.9566
Log-likelihood	NLTK t	0.8355
Log-likelihood	NLTK G^2	0.9886
NLTK PMI	NLTK chi-square	0.9108
NLTK PMI	NLTK t	0.6728
NLTK PMI	NLTK G^2	0.7832
NLTK chi-square	NLTK t	0.7935
NLTK chi-square	NLTK G^2	0.9651
NLTK t	NLTK G^2	0.8307

Every off-diagonal Spearman correlation on the shared filtered bigrams. The heatmap is the same matrix.

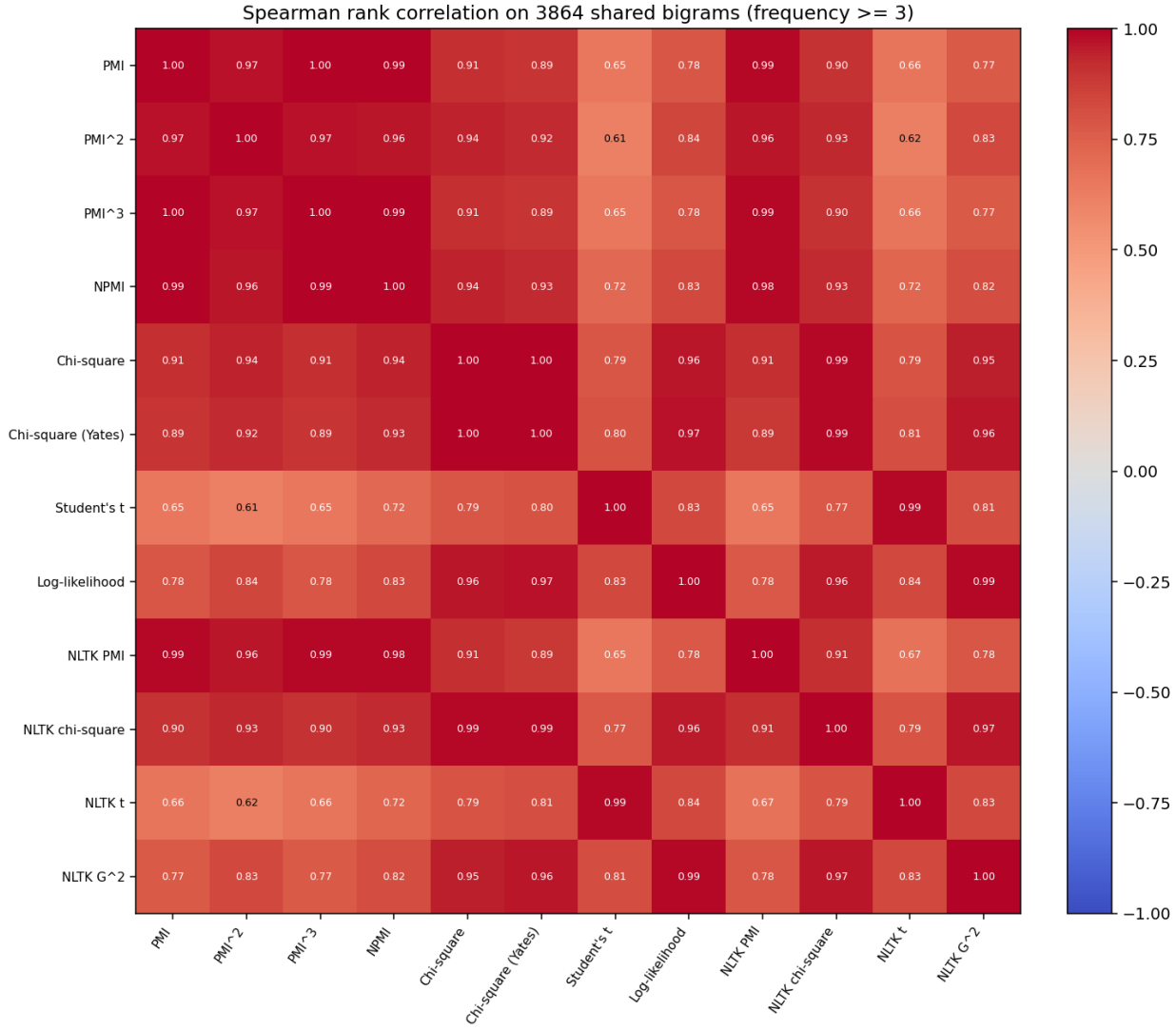


Figure. Spearman correlations. PMI-family scores cluster. The t-test is the outlier.

Top-20 overlap

metric A	metric B	shared top-20	Jaccard
PMI	PMI^2	20	1.000
PMI	PMI^3	20	1.000
PMI	NPMI	14	0.538
PMI	Chi-square	13	0.481
PMI	Chi-square (Yates)	12	0.429
PMI	Student's t	0	0.000
PMI	Log-likelihood	0	0.000
PMI	NLTK PMI	19	0.905
PMI	NLTK chi-square	13	0.481
PMI	NLTK t	0	0.000
PMI	NLTK G^2	0	0.000
PMI^2	PMI^3	20	1.000
PMI^2	NPMI	14	0.538
PMI^2	Chi-square	13	0.481
PMI^2	Chi-square (Yates)	12	0.429
PMI^2	Student's t	0	0.000
PMI^2	Log-likelihood	0	0.000
PMI^2	NLTK PMI	19	0.905
PMI^2	NLTK chi-square	13	0.481
PMI^2	NLTK t	0	0.000
PMI^2	NLTK G^2	0	0.000
PMI^3	NPMI	14	0.538
PMI^3	Chi-square	13	0.481
PMI^3	Chi-square (Yates)	12	0.429
PMI^3	Student's t	0	0.000
PMI^3	Log-likelihood	0	0.000
PMI^3	NLTK PMI	19	0.905
PMI^3	NLTK chi-square	13	0.481
PMI^3	NLTK t	0	0.000
PMI^3	NLTK G^2	0	0.000
NPMI	Chi-square	19	0.905
NPMI	Chi-square (Yates)	18	0.818
NPMI	Student's t	0	0.000
NPMI	Log-likelihood	1	0.026
NPMI	NLTK PMI	13	0.481
NPMI	NLTK chi-square	19	0.905
NPMI	NLTK t	0	0.000
NPMI	NLTK G^2	1	0.026
Chi-square	Chi-square (Yates)	19	0.905
Chi-square	Student's t	0	0.000
Chi-square	Log-likelihood	1	0.026
Chi-square	NLTK PMI	12	0.429
Chi-square	NLTK chi-square	19	0.905
Chi-square	NLTK t	0	0.000
Chi-square	NLTK G^2	1	0.026
Chi-square (Yates)	Student's t	0	0.000
Chi-square (Yates)	Log-likelihood	1	0.026
Chi-square (Yates)	NLTK PMI	11	0.379

metric A	metric B	shared top-20	Jaccard
Chi-square (Yates)	NLTK chi-square	18	0.818
Chi-square (Yates)	NLTK t	0	0.000
Chi-square (Yates)	NLTK G^2	1	0.026
Student's t	Log-likelihood	13	0.481
Student's t	NLTK PMI	0	0.000
Student's t	NLTK chi-square	0	0.000
Student's t	NLTK t	20	1.000
Student's t	NLTK G^2	13	0.481
Log-likelihood	NLTK PMI	0	0.000
Log-likelihood	NLTK chi-square	1	0.026
Log-likelihood	NLTK t	13	0.481
Log-likelihood	NLTK G^2	18	0.818
NLTK PMI	NLTK chi-square	13	0.481
NLTK PMI	NLTK t	0	0.000
NLTK PMI	NLTK G^2	0	0.000
NLTK chi-square	NLTK t	0	0.000
NLTK chi-square	NLTK G^2	1	0.026
NLTK t	NLTK G^2	13	0.481

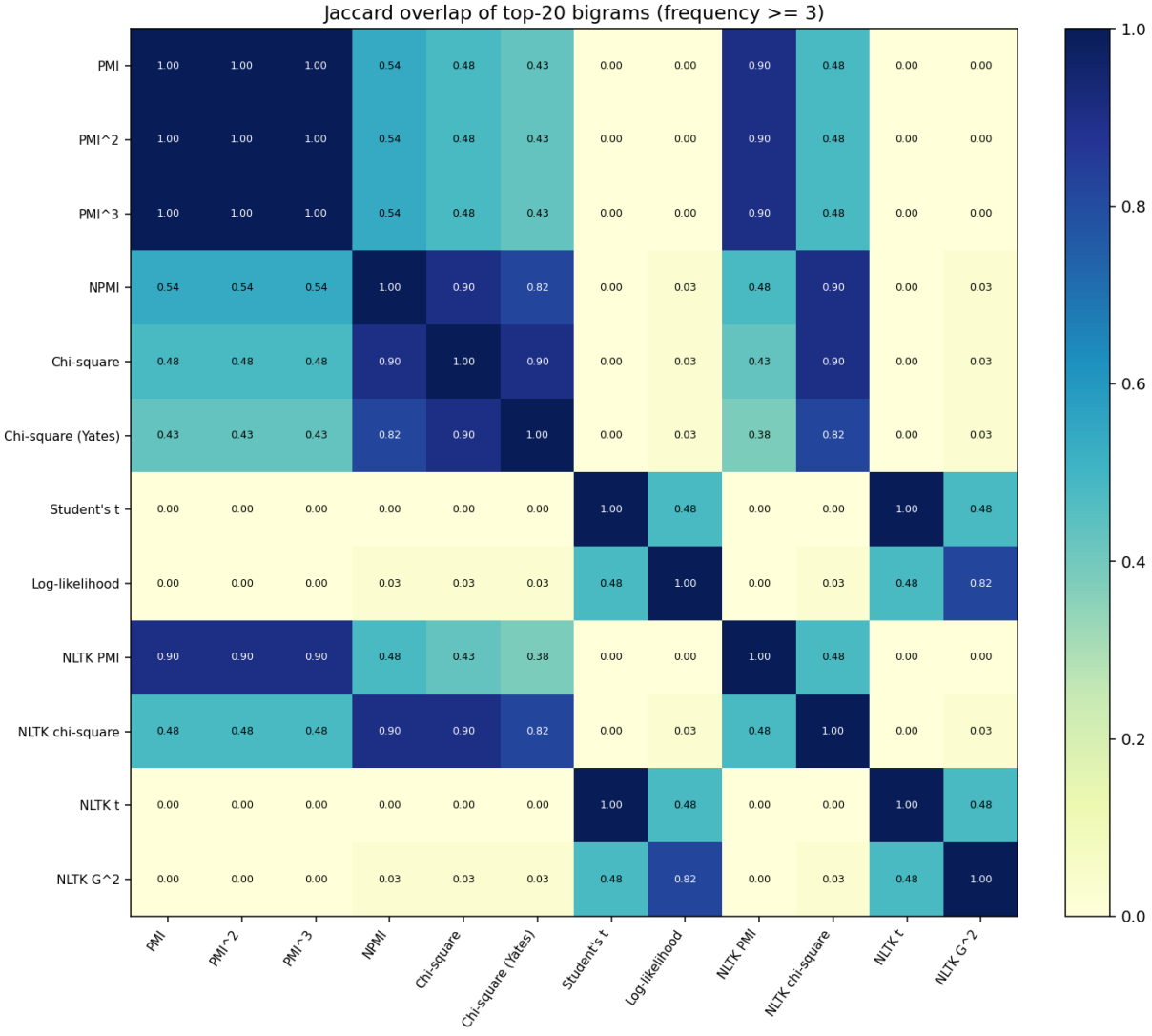


Figure. Jaccard overlap of the filtered top-20 lists.

From-scratch values against NLTK

finder.N = 67046 and our bigram N = 61728. PMI log base decision: log2 (median ratio 1.0000). G^2 log base decision: ln. Chi-square variant: pearson no yates.

metric	largest finder gap	f	ours	NLTK finder	|diff|
PMI	yes thank	3	5.8103	2.7155	3.0949
Chi-square	brook tucker	4	35271.4285	19500.9452	15770.4832
Student's t	the i	4	-27.6966	-39.5325	11.8359
Log-likelihood	and i	271	602.3973	426.3573	176.0400

Same-count differences for chi-square against Pearson are numerically zero at the reported precision. Finder differences are the N convention and the unigram-versus-positional marginals. Full pair tables are in reports/nltk_vs_scratch_*.csv.

Frequency against score

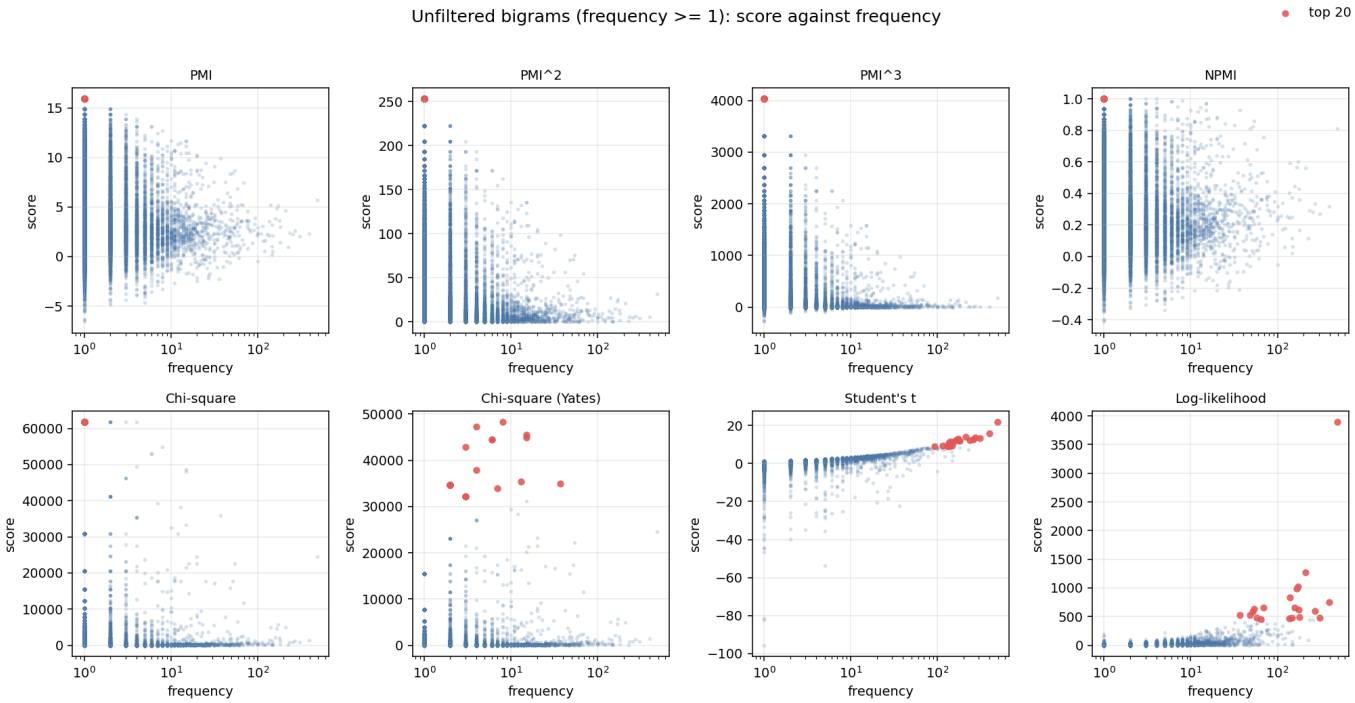


Figure. Every unfiltered bigram. Red points are that metric's own top 20. PMI's red points sit on frequency 1. The t-test's red points sit at the high-frequency edge. Yates is the chi-square panel that does not paint the hapax cloud red.

Unfiltered PMI top, all frequency 1, in alphabetical tie-break order: abstention iraqi, accommodate zuma, alma mater, amilcar cabral, anthony visioning, aristides pereira, arrow miami, artist designer. Several are real names. Several are one-sentence accidents. The tie, not a graded association strength, produced the order.

Qualitative top-20 review

Each row is the filtered ranking (frequency at least 3). genuine means the phrase is on the gold list. acceptable means it is a fixed expression or a real name the gold list did not include. junk means grammar, a lemmatization artifact, a backchannel, or a window cut out of a list. An acceptable miss lowers measured precision and is not an extraction error.

PMI

#	phrase	score	f	label	commentary
1	coffee shop	14.3287	3	genuine	Fixed commercial compound. Count 3 is at or near the frequency floor, and the score 14.3287 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
2	boot camp	13.9136	4	genuine	Fixed program name. The two words do not productively swap. Count 4 is at or near the frequency floor, and the score 13.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
3	ocean boulevard	13.9136	3	genuine	Fixed street name. Count 3 is at or near the frequency floor, and the score 13.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
4	recreation assistant	13.9136	3	genuine	Fixed job title. Count 3 is at or near the frequency floor, and the score 13.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
5	vince lombardi	13.9136	3	genuine	Fixed personal name. Count 3 is at or near the frequency floor, and the score 13.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
6	lombardi trophy	13.5917	4	genuine	Fixed trophy name. Count 4 is at or near the frequency floor, and the score 13.5917 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
7	task force	13.3287	3	genuine	Fixed institutional compound. Count 3 is at or near the frequency floor, and the score 13.3287 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
8	tony frank	13.1767	3	genuine	Fixed personal name. Count 3 is at or near the frequency floor, and the score 13.1767 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
9	aaron scott	13.1063	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Count 4 is at or near the frequency floor, and the score 13.1063 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
10	brook tucker	13.1063	4	genuine	Fixed personal name. Count 4 is at or near the frequency floor, and the score 13.1063 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
11	capitol hill	13.1063	4	genuine	Fixed neighborhood name. Count 4 is at or near the frequency floor, and the score 13.1063 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
12	mm hmm	13.1063	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. PMI still scores it at 13.1063 with count 6 because the two pieces are nearly exclusive. The measure only compares the joint probability with the product of the marginals. It does not know that a backchannel or a grammatical pair is not a tag.
13	pro tem	13.1063	4	genuine	The short form of president pro tempore. Count 4 is at or near the frequency floor, and the score 13.1063 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
14	san francisco	13.1063	6	genuine	City name. Count 6 is at or near the frequency floor, and the score 13.1063 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.

#	phrase	score	f	label	commentary
15	attorney fee	12.9136	3	genuine	Fixed legal-finance compound, lemmatized from 'attorney fees'. Count 3 is at or near the frequency floor, and the score 12.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
16	orange crush	12.9136	3	genuine	Fixed local name (the Orange Crush interchange), not a compositional color-plus-verb. Count 3 is at or near the frequency floor, and the score 12.9136 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
17	core mission	12.7437	3	acceptable	Fixed institutional noun ('core mission') left off the gold list. PMI ranks it with count 3 for the same reason it ranks gold terms: the marginals are small. Low-frequency bias is why this fixed expression outruns 'city council'. Leaving it off the gold list makes the precision@20 hit count a lower bound.
18	dry ice	12.7437	8	genuine	Fixed substance name that happens to be the topic of one hearing. Count 8 is at or near the frequency floor, and the score 12.7437 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
19	santa clara	12.7437	4	genuine	Place name. Count 4 is at or near the frequency floor, and the score 12.7437 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
20	santa cruz	12.7437	3	acceptable	Place name. The gold list kept the longer 'santa cruz park' and not this bigram. PMI ranks it with count 3 for the same reason it ranks gold terms: the marginals are small. Low-frequency bias is why this fixed expression outruns 'city council'. Leaving it off the gold list makes the precision@20 hit count a lower bound.

PMI^2

#	phrase	score	f	label	commentary
1	coffee shop	205.3109	3	genuine	Fixed commercial compound. The underlying PMI is positive, so squaring (score 205.311, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
2	boot camp	193.5893	4	genuine	Fixed program name. The two words do not productively swap. The underlying PMI is positive, so squaring (score 193.589, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
3	ocean boulevard	193.5893	3	genuine	Fixed street name. The underlying PMI is positive, so squaring (score 193.589, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
4	recreation assistant	193.5893	3	genuine	Fixed job title. The underlying PMI is positive, so squaring (score 193.589, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
5	vince lombardi	193.5893	3	genuine	Fixed personal name. The underlying PMI is positive, so squaring (score 193.589, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
6	lombardi trophy	184.7346	4	genuine	Fixed trophy name. The underlying PMI is positive, so squaring (score 184.735, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.

#	phrase	score	f	label	commentary
7	task force	177.6536	3	genuine	Fixed institutional compound. The underlying PMI is positive, so squaring (score 177.654, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
8	tony frank	173.6247	3	genuine	Fixed personal name. The underlying PMI is positive, so squaring (score 173.625, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
9	aaron scott	171.7746	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. The underlying PMI is positive, so squaring (score 171.775, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
10	brook tucker	171.7746	4	genuine	Fixed personal name. The underlying PMI is positive, so squaring (score 171.775, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
11	capitol hill	171.7746	4	genuine	Fixed neighborhood name. The underlying PMI is positive, so squaring (score 171.775, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
12	mm hmm	171.7746	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. The underlying PMI is positive, so squaring (score 171.775, count 6) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
13	pro tem	171.7746	4	genuine	The short form of president pro tempore. The underlying PMI is positive, so squaring (score 171.775, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
14	san francisco	171.7746	6	genuine	City name. The underlying PMI is positive, so squaring (score 171.775, count 6) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
15	attorney fee	166.7620	3	genuine	Fixed legal-finance compound, lemmatized from 'attorney fees'. The underlying PMI is positive, so squaring (score 166.762, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
16	orange crush	166.7620	3	genuine	Fixed local name (the Orange Crush interchange), not a compositional color-plus-verb. The underlying PMI is positive, so squaring (score 166.762, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
17	core mission	162.4022	3	acceptable	Fixed institutional noun ('core mission') left off the gold list. The underlying PMI is positive, so squaring (score 162.402, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.

#	phrase	score	f	label	commentary
18	dry ice	162.4022	8	genuine	Fixed substance name that happens to be the topic of one hearing. The underlying PMI is positive, so squaring (score 162.402, count 8) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
19	santa clara	162.4022	4	genuine	Place name. The underlying PMI is positive, so squaring (score 162.402, count 4) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.
20	santa cruz	162.4022	3	acceptable	Place name. The gold list kept the longer 'santa cruz park' and not this bigram. The underlying PMI is positive, so squaring (score 162.402, count 3) keeps this row in the same place as plain PMI. PMI squared would also promote a large negative PMI such as 'the to', but those squared values stay far below this head. The top 20 therefore copies PMI's low-frequency bias rather than correcting it.

PMI^3

#	phrase	score	f	label	commentary
1	coffee shop	2941.8335	3	genuine	Fixed commercial compound. Cubing a positive PMI (score 2941.83, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
2	boot camp	2693.5314	4	genuine	Fixed program name. The two words do not productively swap. Cubing a positive PMI (score 2693.53, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
3	ocean boulevard	2693.5314	3	genuine	Fixed street name. Cubing a positive PMI (score 2693.53, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
4	recreation assistant	2693.5314	3	genuine	Fixed job title. Cubing a positive PMI (score 2693.53, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
5	vince lombardi	2693.5314	3	genuine	Fixed personal name. Cubing a positive PMI (score 2693.53, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
6	lombardi trophy	2510.8585	4	genuine	Fixed trophy name. Cubing a positive PMI (score 2510.86, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
7	task force	2367.8868	3	genuine	Fixed institutional compound. Cubing a positive PMI (score 2367.89, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
8	tony frank	2287.7954	3	genuine	Fixed personal name. Cubing a positive PMI (score 2287.8, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
9	aaron scott	2251.3270	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Cubing a positive PMI (score 2251.33, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
10	brook tucker	2251.3270	4	genuine	Fixed personal name. Cubing a positive PMI (score 2251.33, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.

#	phrase	score	f	label	commentary
11	capitol hill	2251.3270	4	genuine	Fixed neighborhood name. Cubing a positive PMI (score 2251.33, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
12	mm hmm	2251.3270	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. Cubing a positive PMI (score 2251.33, count 6) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
13	pro tem	2251.3270	4	genuine	The short form of president pro tempore. Cubing a positive PMI (score 2251.33, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
14	san francisco	2251.3270	6	genuine	City name. Cubing a positive PMI (score 2251.33, count 6) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
15	attorney fee	2153.5044	3	genuine	Fixed legal-finance compound, lemmatized from 'attorney fees'. Cubing a positive PMI (score 2153.5, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
16	orange crush	2153.5044	3	genuine	Fixed local name (the Orange Crush interchange), not a compositional color-plus-verb. Cubing a positive PMI (score 2153.5, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
17	core mission	2069.6070	3	acceptable	Fixed institutional noun ('core mission') left off the gold list. Cubing a positive PMI (score 2069.61, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
18	dry ice	2069.6070	8	genuine	Fixed substance name that happens to be the topic of one hearing. Cubing a positive PMI (score 2069.61, count 8) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
19	santa clara	2069.6070	4	genuine	Place name. Cubing a positive PMI (score 2069.61, count 4) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.
20	santa cruz	2069.6070	3	acceptable	Place name. The gold list kept the longer 'santa cruz park' and not this bigram. Cubing a positive PMI (score 2069.61, count 3) is strictly monotone, so this row matches plain PMI exactly. PMI cubed stretches the gaps among rare exclusive pairs and does not bring high-frequency titles up.

NPMI

#	phrase	score	f	label	commentary
1	boot camp	1.0000	4	genuine	Fixed program name. The two words do not productively swap. NPMI score 1 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
2	coffee shop	1.0000	3	genuine	Fixed commercial compound. NPMI score 1 at count 3 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
3	dry ice	0.9868	8	genuine	Fixed substance name that happens to be the topic of one hearing. NPMI score 0.986841 at count 8 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.

#	phrase	score	f	label	commentary
4	mm hmm	0.9833	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. NPMI is essentially 1 when the joint count 6 accounts for both marginals, which is why a backchannel ties the real names. Dividing by -log P(w1, w2) removes raw PMI's extra reward for rarity, and it does not remove exclusivity.
5	san francisco	0.9833	6	genuine	City name. NPMI score 0.983315 at count 6 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
6	lombardi trophy	0.9769	4	genuine	Fixed trophy name. NPMI score 0.976862 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
7	domestic violence	0.9716	15	genuine	Fixed policy term. NPMI score 0.971596 at count 15 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
8	ocean boulevard	0.9710	3	genuine	Fixed street name. NPMI score 0.971034 at count 3 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
9	recreation assistant	0.9710	3	genuine	Fixed job title. NPMI score 0.971034 at count 3 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
10	vince lombardi	0.9710	3	genuine	Fixed personal name. NPMI score 0.971034 at count 3 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
11	super bowl	0.9699	15	genuine	Fixed event name. NPMI score 0.969922 at count 15 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
12	maiden speech	0.9502	7	genuine	Fixed parliamentary formula for a member's first speech. NPMI score 0.950247 at count 7 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
13	cabo verde	0.9433	13	genuine	Country name used as a hearing topic, not a compositional pairing. NPMI score 0.94335 at count 13 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
14	aaron scott	0.9420	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. NPMI score 0.941974 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
15	brook tucker	0.9420	4	genuine	Fixed personal name. NPMI score 0.941974 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
16	capitol hill	0.9420	4	genuine	Fixed neighborhood name. NPMI score 0.941974 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.

#	phrase	score	f	label	commentary
17	pro tem	0.9420	4	genuine	The short form of president pro tempore. NPMI score 0.941974 at count 4 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
18	task force	0.9302	3	genuine	Fixed institutional compound. NPMI score 0.93021 at count 3 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
19	long beach	0.9268	37	genuine	City name, non-compositional. NPMI score 0.926831 at count 37 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.
20	real estate	0.9265	10	genuine	Fixed industry term. NPMI score 0.92646 at count 10 is high because the pair covers its own marginals, not because the count is large. Normalization lets a few higher-count exclusive names ('long beach', 'domestic violence') climb past some frequency-3 pairs that raw PMI had ranked above them.

Chi-square

#	phrase	score	f	label	commentary
1	boot camp	61728.0000	4	genuine	Fixed program name. The two words do not productively swap. Chi-square score 61728 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
2	coffee shop	61728.0000	3	genuine	Fixed commercial compound. Chi-square score 61728 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
3	dry ice	54868.4443	8	genuine	Fixed substance name that happens to be the topic of one hearing. Chi-square score 54868.4 at count 8 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
4	mm hmm	52908.8571	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. Pearson's chi-square is non-directional. Score 52908.9 at count 6 says the cell is dependent, and a backchannel that never occurs apart is as dependent as a policy term. The statistic does not penalize that the way PMI's information ratio is sometimes claimed to.
5	san francisco	52908.8571	6	genuine	City name. Chi-square score 52908.9 at count 6 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
6	lombardi trophy	49381.5999	4	genuine	Fixed trophy name. Chi-square score 49381.6 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
7	domestic violence	48729.4729	15	genuine	Fixed policy term. Chi-square score 48729.5 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
8	super bowl	48054.6012	15	genuine	Fixed event name. Chi-square score 48054.6 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.

#	phrase	score	f	label	commentary
9	ocean boulevard	46295.2500	3	genuine	Fixed street name. Chi-square score 46295.2 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
10	recreation assistant	46295.2500	3	genuine	Fixed job title. Chi-square score 46295.2 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
11	vince lombardi	46295.2500	3	genuine	Fixed personal name. Chi-square score 46295.2 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
12	maiden speech	39278.9088	7	genuine	Fixed parliamentary formula for a member's first speech. Chi-square score 39278.9 at count 7 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
13	cabo verde	38207.6180	13	genuine	Country name used as a hearing topic, not a compositional pairing. Chi-square score 38207.6 at count 13 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
14	long beach	35852.3621	37	genuine	City name, non-compositional. Chi-square score 35852.4 at count 37 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
15	aaron scott	35271.4285	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Chi-square score 35271.4 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
16	brook tucker	35271.4285	4	genuine	Fixed personal name. Chi-square score 35271.4 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
17	capitol hill	35271.4285	4	genuine	Fixed neighborhood name. Chi-square score 35271.4 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
18	pro tem	35271.4285	4	genuine	The short form of president pro tempore. Chi-square score 35271.4 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
19	solid waste	33379.2529	15	genuine	Fixed utility term. Chi-square score 33379.3 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
20	real estate	32483.6834	10	genuine	Fixed industry term. Chi-square score 32483.7 at count 10 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.

Chi-square (Yates)

#	phrase	score	f	label	commentary
1	dry ice	48223.2810	8	genuine	Fixed substance name that happens to be the topic of one hearing. Yates subtracts N/2 inside the absolute deviation before squaring, so score 48223.3 at count 8 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
2	boot camp	47259.6249	4	genuine	Fixed program name. The two words do not productively swap. Yates subtracts N/2 inside the absolute deviation before squaring, so score 47259.6 at count 4 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
3	domestic violence	45534.0184	15	genuine	Fixed policy term. Yates subtracts N/2 inside the absolute deviation before squaring, so score 45534 at count 15 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
4	super bowl	44903.3882	15	genuine	Fixed event name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 44903.4 at count 15 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
5	mm hmm	44457.2201	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. Yates subtracts N/2 inside the absolute deviation before squaring, so score 44457.2 at count 6 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
6	san francisco	44457.2201	6	genuine	City name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 44457.2 at count 6 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
7	coffee shop	42865.8333	3	genuine	Fixed commercial compound. Yates subtracts N/2 inside the absolute deviation before squaring, so score 42865.8 at count 3 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
8	lombardi trophy	37806.9124	4	genuine	Fixed trophy name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 37806.9 at count 4 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
9	cabo verde	35324.1289	13	genuine	Country name used as a hearing topic, not a compositional pairing. Yates subtracts N/2 inside the absolute deviation before squaring, so score 35324.1 at count 13 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
10	long beach	34888.9394	37	genuine	City name, non-compositional. Yates subtracts N/2 inside the absolute deviation before squaring, so score 34888.9 at count 37 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.

#	phrase	score	f	label	commentary
11	maiden speech	33867.1100	7	genuine	Fixed parliamentary formula for a member's first speech. Yates subtracts N/2 inside the absolute deviation before squaring, so score 33867.1 at count 7 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
12	ocean boulevard	32148.6458	3	genuine	Fixed street name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 32148.6 at count 3 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
13	recreation assistant	32148.6458	3	genuine	Fixed job title. Yates subtracts N/2 inside the absolute deviation before squaring, so score 32148.6 at count 3 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
14	vince lombardi	32148.6458	3	genuine	Fixed personal name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 32148.6 at count 3 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
15	solid waste	31190.0906	15	genuine	Fixed utility term. Yates subtracts N/2 inside the absolute deviation before squaring, so score 31190.1 at count 15 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
16	real estate	29315.5742	10	genuine	Fixed industry term. Yates subtracts N/2 inside the absolute deviation before squaring, so score 29315.6 at count 10 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
17	cape verdean	28339.1133	12	acceptable	Demonym for the Cabo Verde community. Real, and not on the gold list, which kept the place name instead. Yates subtracts N/2 inside the absolute deviation before squaring, so score 28339.1 at count 12 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term. This is the Yates-only phrase in the eight from-scratch top-20 lists.
18	aaron scott	27003.8124	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Yates subtracts N/2 inside the absolute deviation before squaring, so score 27003.8 at count 4 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
19	brook tucker	27003.8124	4	genuine	Fixed personal name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 27003.8 at count 4 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.
20	capitol hill	27003.8124	4	genuine	Fixed neighborhood name. Yates subtracts N/2 inside the absolute deviation before squaring, so score 27003.8 at count 4 is pulled down relative to the uncorrected chi-square, more so for smaller counts. The correction breaks the saturated tie at N and prefers exclusive pairs that have a bit more mass. It is still non-directional, so a backchannel remains competitive with a policy term.

Student's t

#	phrase	score	f	label	commentary
1	thank you	21.6569	488	acceptable	The dominant politeness formula in council speech. Fixed, and not a domain tag. Student's t score 21.6569 is large because the count is 488, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
2	of the	15.8955	396	junk	The highest-frequency grammatical bigram. Dependence here is English, not a domain. Student's t score 15.8955 is large because the count is 396, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
3	want to	13.9023	210	junk	Verb plus infinitive marker. It combines with anything. Student's t score 13.9023 is large because the count is 210, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
4	and i	13.7651	271	junk	Coordinator plus first-person pronoun. It is syntax, not a term. Student's t score 13.7651 is large because the count is 271, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
5	i be	13.3498	306	junk	Lemmatized 'I am'. A copula artifact, not a collocation. Student's t score 13.3498 is large because the count is 306, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
6	you know	12.7310	172	acceptable	A fixed discourse marker. Not a domain tag. Student's t score 12.731 is large because the count is 172, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
7	in the	12.6011	256	junk	Preposition plus determiner, the prototype of grammatical co-occurrence. Student's t score 12.6011 is large because the count is 256, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
8	i think	12.4726	167	junk	A stance marker. Speakers use it with any complement. Student's t score 12.4726 is large because the count is 167, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
9	we be	12.3145	234	junk	Lemmatized 'we are'. Student's t score 12.3145 is large because the count is 234, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
10	we have	12.2191	177	junk	Pronoun plus verb, no lexical term. Student's t score 12.2191 is large because the count is 177, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
11	that we	11.7771	180	junk	Complementizer plus pronoun. Student's t score 11.7771 is large because the count is 180, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.

#	phrase	score	f	label	commentary
12	go to	11.7396	159	junk	A high-frequency verb plus particle. Student's t score 11.7396 is large because the count is 159, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
13	do not	11.5161	140	junk	Clitic repair of 'don't' plus the auxiliary. Grammatical, not topical. Student's t score 11.5161 is large because the count is 140, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
14	the city	10.8072	146	acceptable	A compositional frame that is constant in this genre. Not gold, because 'city council' and 'city of' carry the name. Student's t score 10.8072 is large because the count is 146, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
15	so i	10.6857	136	junk	Discourse glue. Student's t score 10.6857 is large because the count is 136, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
16	this be	9.4270	148	junk	Lemmatized 'this is'. The tagger created a collocation that was never spoken. Student's t score 9.42702 is large because the count is 148, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators. Among the from-scratch metrics, only the t-test promotes 'this be' into the top 20.
17	there be	9.4240	114	junk	Lemmatized 'there is' / 'there are'. Student's t score 9.42396 is large because the count is 114, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
18	it be	9.0718	137	junk	Lemmatized 'it is'. Student's t score 9.07176 is large because the count is 137, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
19	i would	8.9885	93	junk	Pronoun plus modal. No lexical cohesion. Student's t score 8.9885 is large because the count is 93, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
20	on the	8.9855	130	junk	Preposition plus determiner. Student's t score 8.98555 is large because the count is 130, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.

Log-likelihood

#	phrase	score	f	label	commentary
1	thank you	3890.2927	488	acceptable	The dominant politeness formula in council speech. Fixed, and not a domain tag. G^2 score 3890.29 at count 488 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
2	want to	1274.0054	210	junk	Verb plus infinitive marker. It combines with anything. G^2 score 1274.01 at count 210 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.

#	phrase	score	f	label	commentary
3	you know	1025.2401	172	acceptable	A fixed discourse marker. Not a domain tag. G^2 score 1025.24 at count 172 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
4	i think	988.9872	167	junk	A stance marker. Speakers use it with any complement. G^2 score 988.987 at count 167 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
5	do not	829.2815	140	junk	Clitic repair of 'don't' plus the auxiliary. Grammatical, not topical. G^2 score 829.281 at count 140 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
6	of the	745.7410	396	junk	The highest-frequency grammatical bigram. Dependence here is English, not a domain. G^2 score 745.741 at count 396 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
7	go to	660.9346	159	junk	A high-frequency verb plus particle. G^2 score 660.935 at count 159 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
8	council member	651.6960	69	genuine	The standard role title in this corpus. G^2 score 651.696 at count 69 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface. No other from-scratch metric places 'council member' in its top 20.
9	as well	630.0289	54	acceptable	A fixed additive discourse formula, deliberately not a domain tag. G^2 score 630.029 at count 54 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
10	we have	623.3646	177	junk	Pronoun plus verb, no lexical term. G^2 score 623.365 at count 177 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
11	and i	602.3973	271	junk	Coordinator plus first-person pronoun. It is syntax, not a term. G^2 score 602.397 at count 271 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
12	make sure	585.9251	52	acceptable	A fixed discourse verb. Not a topic tag. G^2 score 585.925 at count 52 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
13	long beach	530.1656	37	genuine	City name, non-compositional. G^2 score 530.166 at count 37 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface.
14	mr president	521.9213	48	genuine	Fixed form of address to the chair. G^2 score 521.921 at count 48 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface.
15	that we	490.1779	180	junk	Complementizer plus pronoun. G^2 score 490.178 at count 180 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.

#	phrase	score	f	label	commentary
16	look at	477.8333	58	junk	Verb plus particle, freely composed. G^2 score 477.833 at count 58 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
17	the city	477.8134	146	acceptable	A compositional frame that is constant in this genre. Not gold, because 'city council' and 'city of' carry the name. G^2 score 477.813 at count 146 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
18	i be	473.3965	306	junk	Lemmatized 'I am'. A copula artifact, not a collocation. G^2 score 473.396 at count 306 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
19	so i	467.0963	136	junk	Discourse glue. G^2 score 467.096 at count 136 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
20	kind of	455.1421	64	acceptable	A fixed hedge. Not a domain tag, and not random. G^2 score 455.142 at count 64 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.

NLTK PMI

#	phrase	score	f	label	commentary
1	coffee shop	14.4479	3	genuine	Fixed commercial compound. Count 3 is at or near the frequency floor, and the score 14.4479 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
2	boot camp	14.0329	4	genuine	Fixed program name. The two words do not productively swap. Count 4 is at or near the frequency floor, and the score 14.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
3	ocean boulevard	14.0329	3	genuine	Fixed street name. Count 3 is at or near the frequency floor, and the score 14.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
4	recreation assistant	14.0329	3	genuine	Fixed job title. Count 3 is at or near the frequency floor, and the score 14.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
5	vince lombardi	14.0329	3	genuine	Fixed personal name. Count 3 is at or near the frequency floor, and the score 14.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
6	lombardi trophy	13.7109	4	genuine	Fixed trophy name. Count 4 is at or near the frequency floor, and the score 13.7109 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
7	task force	13.4479	3	genuine	Fixed institutional compound. Count 3 is at or near the frequency floor, and the score 13.4479 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
8	tony frank	13.2959	3	genuine	Fixed personal name. Count 3 is at or near the frequency floor, and the score 13.2959 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.

#	phrase	score	f	label	commentary
9	aaron scott	13.2255	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Count 4 is at or near the frequency floor, and the score 13.2255 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
10	capitol hill	13.2255	4	genuine	Fixed neighborhood name. Count 4 is at or near the frequency floor, and the score 13.2255 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
11	pro tem	13.2255	4	genuine	The short form of president pro tempore. Count 4 is at or near the frequency floor, and the score 13.2255 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
12	mm hmm	13.2255	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. PMI still scores it at 13.2255 with count 6 because the two pieces are nearly exclusive. The measure only compares the joint probability with the product of the marginals. It does not know that a backchannel or a grammatical pair is not a tag.
13	attorney fee	13.0329	3	genuine	Fixed legal-finance compound, lemmatized from 'attorney fees'. Count 3 is at or near the frequency floor, and the score 13.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
14	san francisco	13.0329	6	genuine	City name. Count 6 is at or near the frequency floor, and the score 13.0329 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
15	dry ice	12.8629	8	genuine	Fixed substance name that happens to be the topic of one hearing. Count 8 is at or near the frequency floor, and the score 12.8629 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
16	santa clara	12.8629	4	genuine	Place name. Count 4 is at or near the frequency floor, and the score 12.8629 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
17	core mission	12.8629	3	acceptable	Fixed institutional noun ('core mission') left off the gold list. PMI ranks it with count 3 for the same reason it ranks gold terms: the marginals are small. Low-frequency bias is why this fixed expression outruns 'city council'. Leaving it off the gold list makes the precision@20 hit count a lower bound.
18	orange crush	12.8629	3	genuine	Fixed local name (the Orange Crush interchange), not a compositional color-plus-verb. Count 3 is at or near the frequency floor, and the score 12.8629 is high because each word barely occurs outside the pair. That is PMI's low-frequency bias. A frequent title such as 'council member' is more useful to a tagger and ranks lower.
19	santa cruz	12.8629	3	acceptable	Place name. The gold list kept the longer 'santa cruz park' and not this bigram. PMI ranks it with count 3 for the same reason it ranks gold terms: the marginals are small. Low-frequency bias is why this fixed expression outruns 'city council'. Leaving it off the gold list makes the precision@20 hit count a lower bound.
20	tara rise	12.8629	3	acceptable	A street or subdivision name that NLTK's marginals ranked into the PMI top 20. Not gold-listed. PMI ranks it with count 3 for the same reason it ranks gold terms: the marginals are small. Low-frequency bias is why this fixed expression outruns 'city council'. Leaving it off the gold list makes the precision@20 hit count a lower bound. Our PMI ranks 'brook tucker' in this slot. NLTK's unigram marginals and finder.N, which counts tokens rather than bigrams, swap the two.

NLTK chi-square

#	phrase	score	f	label	commentary
1	boot camp	67046.0000	4	genuine	Fixed program name. The two words do not productively swap. Chi-square score 67046 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
2	coffee shop	67046.0000	3	genuine	Fixed commercial compound. Chi-square score 67046 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
3	dry ice	59595.5554	8	genuine	Fixed substance name that happens to be the topic of one hearing. Chi-square score 59595.6 at count 8 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
4	mm hmm	57467.1428	6	junk	A backchannel. The two syllables almost never occur apart, so dependence scores treat a noise token as a term. Pearson's chi-square is non-directional. Score 57467.1 at count 6 says the cell is dependent, and a backchannel that never occurs apart is as dependent as a policy term. The statistic does not penalize that the way PMI's information ratio is sometimes claimed to.
5	lombardi trophy	53636.0000	4	genuine	Fixed trophy name. Chi-square score 53636 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
6	domestic violence	52927.8940	15	genuine	Fixed policy term. Chi-square score 52927.9 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
7	super bowl	52194.9127	15	genuine	Fixed event name. Chi-square score 52194.9 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
8	ocean boulevard	50283.7500	3	genuine	Fixed street name. Chi-square score 50283.7 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
9	recreation assistant	50283.7500	3	genuine	Fixed job title. Chi-square score 50283.7 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
10	vince lombardi	50283.7500	3	genuine	Fixed personal name. Chi-square score 50283.7 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
11	san francisco	50282.9999	6	genuine	City name. Chi-square score 50283 at count 6 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
12	maiden speech	42663.0906	7	genuine	Fixed parliamentary formula for a member's first speech. Chi-square score 42663.1 at count 7 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
13	cabo verde	41499.7133	13	genuine	Country name used as a hearing topic, not a compositional pairing. Chi-square score 41499.7 at count 13 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.

#	phrase	score	f	label	commentary
14	aaron scott	38310.2856	4	genuine	Fixed personal name (given name plus surname) from the meeting floor. Chi-square score 38310.3 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
15	capitol hill	38310.2856	4	genuine	Fixed neighborhood name. Chi-square score 38310.3 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
16	pro tem	38310.2856	4	genuine	The short form of president pro tempore. Chi-square score 38310.3 at count 4 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
17	long beach	37724.3508	37	genuine	City name, non-compositional. Chi-square score 37724.4 at count 37 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
18	solid waste	33664.7526	15	genuine	Fixed utility term. Chi-square score 33664.8 at count 15 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
19	task force	33521.4999	3	genuine	Fixed institutional compound. Chi-square score 33521.5 at count 3 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.
20	real estate	33517.9993	10	genuine	Fixed industry term. Chi-square score 33518 at count 10 is large because the off-diagonal cells are nearly empty, so the pair is far from independence. Exclusive pairs of any count saturate near N. Chi-square therefore piles rare names at the top and does not rank by how frequent a useful tag is.

NLTK t

#	phrase	score	f	label	commentary
1	thank you	21.6533	488	acceptable	The dominant politeness formula in council speech. Fixed, and not a domain tag. Student's t score 21.6533 is large because the count is 488, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
2	of the	15.9321	396	junk	The highest-frequency grammatical bigram. Dependence here is English, not a domain. Student's t score 15.9321 is large because the count is 396, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
3	want to	13.9348	210	junk	Verb plus infinitive marker. It combines with anything. Student's t score 13.9348 is large because the count is 210, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
4	i be	13.5355	306	junk	Lemmatized 'I am'. A copula artifact, not a collocation. Student's t score 13.5355 is large because the count is 306, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.

#	phrase	score	f	label	commentary
5	you know	12.6739	172	acceptable	A fixed discourse marker. Not a domain tag. Student's t score 12.6739 is large because the count is 172, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
6	and i	12.6728	271	junk	Coordinator plus first-person pronoun. It is syntax, not a term. Student's t score 12.6728 is large because the count is 271, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
7	in the	12.6256	256	junk	Preposition plus determiner, the prototype of grammatical co-occurrence. Student's t score 12.6256 is large because the count is 256, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
8	i think	12.4906	167	junk	A stance marker. Speakers use it with any complement. Student's t score 12.4906 is large because the count is 167, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
9	we be	12.4868	234	junk	Lemmatized 'we are'. Student's t score 12.4868 is large because the count is 234, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
10	we have	12.2890	177	junk	Pronoun plus verb, no lexical term. Student's t score 12.289 is large because the count is 177, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
11	go to	11.7713	159	junk	A high-frequency verb plus particle. Student's t score 11.7713 is large because the count is 159, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
12	do not	11.5132	140	junk	Clitic repair of 'don't' plus the auxiliary. Grammatical, not topical. Student's t score 11.5132 is large because the count is 140, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
13	that we	11.4614	180	junk	Complementizer plus pronoun. Student's t score 11.4614 is large because the count is 180, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
14	the city	10.8965	146	acceptable	A compositional frame that is constant in this genre. Not gold, because 'city council' and 'city of' carry the name. Student's t score 10.8965 is large because the count is 146, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
15	so i	10.2674	136	junk	Discourse glue. Student's t score 10.2674 is large because the count is 136, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.

#	phrase	score	f	label	commentary
16	there be	9.3741	114	junk	Lemmatized 'there is' / 'there are'. Student's t score 9.37412 is large because the count is 114, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
17	this be	9.3702	148	junk	Lemmatized 'this is'. The tagger created a collocation that was never spoken. Student's t score 9.37019 is large because the count is 148, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
18	i would	9.0055	93	junk	Pronoun plus modal. No lexical cohesion. Student's t score 9.00552 is large because the count is 93, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
19	on the	8.9836	130	junk	Preposition plus determiner. Student's t score 8.98359 is large because the count is 130, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.
20	it be	8.9813	137	junk	Lemmatized 'it is'. Student's t score 8.98134 is large because the count is 137, not because the pair is a term. The Manning and Schütze form divides a frequency-scale numerator by sqrt(f), so grammatical bigrams outrank domain names. The normal sampling model behind a t statistic is not satisfied by these sparse indicators.

NLTK G^2

#	phrase	score	f	label	commentary
1	thank you	3845.4232	488	acceptable	The dominant politeness formula in council speech. Fixed, and not a domain tag. G^2 score 3845.42 at count 488 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
2	want to	1285.5940	210	junk	Verb plus infinitive marker. It combines with anything. G^2 score 1285.59 at count 210 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
3	i think	996.0715	167	junk	A stance marker. Speakers use it with any complement. G^2 score 996.072 at count 167 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
4	you know	970.1596	172	acceptable	A fixed discourse marker. Not a domain tag. G^2 score 970.16 at count 172 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
5	do not	822.5380	140	junk	Clitic repair of 'don't' plus the auxiliary. Grammatical, not topical. G^2 score 822.538 at count 140 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
6	of the	748.8335	396	junk	The highest-frequency grammatical bigram. Dependence here is English, not a domain. G^2 score 748.834 at count 396 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.

#	phrase	score	f	label	commentary
7	go to	667.7069	159	junk	A high-frequency verb plus particle. G^2 score 667.707 at count 159 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
8	we have	645.1276	177	junk	Pronoun plus verb, no lexical term. G^2 score 645.128 at count 177 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
9	council member	622.5434	69	genuine	The standard role title in this corpus. G^2 score 622.543 at count 69 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface.
10	as well	601.6796	54	acceptable	A fixed additive discourse formula, deliberately not a domain tag. G^2 score 601.68 at count 54 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
11	make sure	553.3945	52	acceptable	A fixed discourse verb. Not a topic tag. G^2 score 553.394 at count 52 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
12	long beach	532.7418	37	genuine	City name, non-compositional. G^2 score 532.742 at count 37 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface.
13	the city	497.5654	146	acceptable	A compositional frame that is constant in this genre. Not gold, because 'city council' and 'city of' carry the name. G^2 score 497.565 at count 146 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
14	i be	495.2625	306	junk	Lemmatized 'I am'. A copula artifact, not a collocation. G^2 score 495.263 at count 306 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
15	mr president	486.9835	48	genuine	Fixed form of address to the chair. G^2 score 486.983 at count 48 is one of the few domain hits in this top 20. The likelihood ratio still grows with sample size and is non-directional, but it is less hypnotized by frequency-3 exclusivity than PMI or Pearson's chi-square, so a repeated title can surface.
16	look at	483.4718	58	junk	Verb plus particle, freely composed. G^2 score 483.472 at count 58 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
17	kind of	464.8858	64	acceptable	A fixed hedge. Not a domain tag, and not random. G^2 score 464.886 at count 64 rewards a repeated fixed formula. Dunning's statistic behaves better than chi-square at low counts, and on this list the mass is not low: high-frequency discourse still fills most of the top 20.
18	we be	464.7240	234	junk	Lemmatized 'we are'. G^2 score 464.724 at count 234 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.
19	in the	448.9079	256	junk	Preposition plus determiner, the prototype of grammatical co-occurrence. G^2 score 448.908 at count 256 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.

#	phrase	score	f	label	commentary
20	and i	426.3573	271	junk	Coordinator plus first-person pronoun. It is syntax, not a term. G^2 score 426.357 at count 271 is large because the observed cell is far from the independence expected count and the count itself is high. The statistic does not point toward association rather than dissociation, and it does not prefer a rare technical term over 'of the'.

Trigram PMI

#	phrase	score	f	label	commentary
1	vince lombardi trophy	27.2682	3	genuine	Fixed trophy name, the trigram that contains the bigram 'lombardi trophy'. Trigram PMI score 27.2682 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
2	wastewater treatment division	25.2503	3	genuine	Fixed department name. Trigram PMI score 25.2503 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
3	president pro tem	23.9248	4	genuine	Fixed officer title. Trigram PMI score 23.9248 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
4	santa cruz park	23.4202	3	genuine	Fixed park name. Trigram PMI score 23.4202 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
5	maximum building height	23.3613	4	genuine	Fixed zoning measure. Trigram PMI score 23.3613 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
6	real estate master	22.7828	3	genuine	The start of a master-plan title. Kept on the gold list as that name. Trigram PMI score 22.7828 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
7	solid waste advisory	22.6933	3	genuine	Fixed committee name, truncated before 'committee' but still the conventional start. Trigram PMI score 22.6933 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
8	housing task force	22.5958	3	genuine	Fixed body name. Trigram PMI score 22.5958 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
9	vice mayor lowenthal	22.4974	3	genuine	Title plus surname, the way the clerk addresses that member. Trigram PMI score 22.4974 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
10	law enforcement oversight	22.4646	7	genuine	Fixed oversight-policy trigram. Trigram PMI score 22.4646 at count 7 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
11	low scale residential	22.3454	3	genuine	Fixed zoning description. Trigram PMI score 22.3454 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.

#	phrase	score	f	label	commentary
12	texas historical commission	22.2503	3	genuine	Fixed agency name. Trigram PMI score 22.2503 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
13	mile high stadium	22.0863	5	genuine	Fixed stadium name. Trigram PMI score 22.0863 at count 5 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
14	utility economic development	21.8088	3	genuine	Gold-listed as the tail of the utilities and economic-development committee name. It is institutional, and it is also a window cut out of a longer list. Trigram PMI score 21.8088 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
15	urban renewal area	21.7198	3	genuine	Fixed planning designation. Trigram PMI score 21.7198 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
16	national restaurant stock	21.5540	3	junk	A damaged slice around the National Western Stock Show, not a name anyone uses. Trigram PMI score 21.554 at count 3 treats a three-token window as a unit. Committee lists and ASR splices produce exclusive windows that are not phrases. The same low-frequency bias as bigram PMI applies, with even sparser counts.
17	favor please signify	21.3387	3	acceptable	A slice of the voting formula 'in favor please signify'. Procedural, and not the trigram that was gold-listed. Trigram PMI score 21.3387 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
18	right utility economic	21.3277	3	junk	The middle of 'Civil Rights, Utilities, Economic Development'. The comma was tokenized away. Trigram PMI score 21.3277 at count 3 treats a three-token window as a unit. Committee lists and ASR splices produce exclusive windows that are not phrases. The same low-frequency bias as bigram PMI applies, with even sparser counts.
19	human relation board	21.2909	4	genuine	Fixed board name, with 'relations' lemmatized to 'relation'. Trigram PMI score 21.2909 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
20	civil right utility	20.9828	3	junk	A slice of the committee title 'Civil Rights, Utilities, Economic Development' after the comma disappeared and 'rights' was lemmatized. Trigram PMI score 20.9828 at count 3 treats a three-token window as a unit. Committee lists and ASR splices produce exclusive windows that are not phrases. The same low-frequency bias as bigram PMI applies, with even sparser counts.

NLTK trigram PMI

#	phrase	score	f	label	commentary
1	vince lombardi trophy	27.7438	3	genuine	Fixed trophy name, the trigram that contains the bigram 'lombardi trophy'. Trigram PMI score 27.7438 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
2	wastewater treatment division	25.7259	3	genuine	Fixed department name. Trigram PMI score 25.7259 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.

#	phrase	score	f	label	commentary
3	santa cruz park	23.8958	3	genuine	Fixed park name. Trigram PMI score 23.8958 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
4	maximum building height	23.5150	4	genuine	Fixed zoning measure. Trigram PMI score 23.515 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
5	president pro tem	22.8661	4	genuine	Fixed officer title. Trigram PMI score 22.8661 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
6	housing task force	22.8369	3	genuine	Fixed body name. Trigram PMI score 22.8369 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
7	law enforcement oversight	22.6057	7	genuine	Fixed oversight-policy trigram. Trigram PMI score 22.6057 at count 7 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
8	low scale residential	22.5579	3	genuine	Fixed zoning description. Trigram PMI score 22.5579 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
9	solid waste advisory	22.5214	3	genuine	Fixed committee name, truncated before 'committee' but still the conventional start. Trigram PMI score 22.5214 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
10	real estate master	22.4219	3	genuine	The start of a master-plan title. Kept on the gold list as that name. Trigram PMI score 22.4219 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
11	texas historical commission	22.3108	3	genuine	Fixed agency name. Trigram PMI score 22.3108 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
12	urban renewal area	22.1770	3	genuine	Fixed planning designation. Trigram PMI score 22.177 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
13	mile high stadium	22.0104	5	genuine	Fixed stadium name. Trigram PMI score 22.0104 at count 5 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
14	vice mayor lowenthal	21.9109	3	genuine	Title plus surname, the way the clerk addresses that member. Trigram PMI score 21.9109 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
15	utility economic development	21.8661	3	genuine	Gold-listed as the tail of the utilities and economic-development committee name. It is institutional, and it is also a window cut out of a longer list. Trigram PMI score 21.8661 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.

#	phrase	score	f	label	commentary
16	human relation board	21.6863	4	genuine	Fixed board name, with 'relations' lemmatized to 'relation'. Trigram PMI score 21.6863 at count 4 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
17	national restaurant stock	21.6437	3	junk	A damaged slice around the National Western Stock Show, not a name anyone uses. Trigram PMI score 21.6437 at count 3 treats a three-token window as a unit. Committee lists and ASR splices produce exclusive windows that are not phrases. The same low-frequency bias as bigram PMI applies, with even sparser counts.
18	annual action plan	20.8490	3	acceptable	A real planning document name (annual action plan) that was not put on the gold list. Trigram PMI score 20.849 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.
19	restaurant stock show	20.8363	3	junk	Another broken window on the stock-show name, with 'restaurant' intruding. Trigram PMI score 20.8363 at count 3 treats a three-token window as a unit. Committee lists and ASR splices produce exclusive windows that are not phrases. The same low-frequency bias as bigram PMI applies, with even sparser counts.
20	estate master plan	20.6563	3	acceptable	The tail of 'real estate master plan'. The full name is longer than this trigram window. Trigram PMI score 20.6563 at count 3 is high because the three position-specific marginals are close to the joint count. Real names survive, and so do fragments of longer names. The frequency floor of 3 is the only thing keeping hapax trigrams out of this list.

NLTK trigram chi-square

#	phrase	score	f	label	commentary
1	vince lombardi trophy	674278272.0496	3	genuine	Fixed trophy name, the trigram that contains the bigram 'lombardi trophy'. NLTK trigram chi-square score 6.74278e+08 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
2	wastewater treatment division	166487632.9259	3	genuine	Fixed department name. NLTK trigram chi-square score 1.66488e+08 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
3	maximum building height	47949553.7273	4	genuine	Fixed zoning measure. NLTK trigram chi-square score 4.79496e+07 at count 4 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
4	santa cruz park	46824645.3241	3	genuine	Fixed park name. NLTK trigram chi-square score 4.68246e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
5	law enforcement oversight	44679042.4208	7	genuine	Fixed oversight-policy trigram. NLTK trigram chi-square score 4.4679e+07 at count 7 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
6	president pro tem	30579359.8105	4	genuine	Fixed officer title. NLTK trigram chi-square score 3.05794e+07 at count 4 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
7	housing task force	22475829.1718	3	genuine	Fixed body name. NLTK trigram chi-square score 2.24758e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
8	mile high stadium	21125028.0818	5	genuine	Fixed stadium name. NLTK trigram chi-square score 2.1125e+07 at count 5 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.

#	phrase	score	f	label	commentary
9	low scale residential	18525627.8436	3	genuine	Fixed zoning description. NLTK trigram chi-square score 1.85256e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
10	solid waste advisory	18082481.2903	3	genuine	Fixed committee name, truncated before 'committee' but still the conventional start. NLTK trigram chi-square score 1.80825e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
11	real estate master	16873297.8944	3	genuine	The start of a master-plan title. Kept on the gold list as that name. NLTK trigram chi-square score 1.68733e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
12	texas historical commission	15608211.8916	3	genuine	Fixed agency name. NLTK trigram chi-square score 1.56082e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
13	urban renewal area	14225487.5381	3	genuine	Fixed planning designation. NLTK trigram chi-square score 1.42255e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
14	human relation board	13498992.5599	4	genuine	Fixed board name, with 'relations' lemmatized to 'relation'. NLTK trigram chi-square score 1.3499e+07 at count 4 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
15	vice mayor lowenthal	11839312.9722	3	genuine	Title plus surname, the way the clerk addresses that member. NLTK trigram chi-square score 1.18393e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
16	utility economic development	11472840.0391	3	genuine	Gold-listed as the tail of the utilities and economic-development committee name. It is institutional, and it is also a window cut out of a longer list. NLTK trigram chi-square score 1.14728e+07 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
17	national restaurant stock	9833175.7489	3	junk	A damaged slice around the National Western Stock Show, not a name anyone uses. NLTK trigram chi-square score 9.83318e+06 at count 3 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
18	cfd dash two	8310581.0254	11	junk	An agenda code tokenized into three pieces ('CFD', the word 'dash', and 'two'). Not a phrase. NLTK trigram chi-square score 8.31058e+06 at count 11 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
19	western stock show	7501586.7716	6	genuine	Fixed event name (National Western Stock Show), the portion the gold list kept. NLTK trigram chi-square score 7.50159e+06 at count 6 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.
20	national western stock	5837326.9848	4	acceptable	The front of 'National Western Stock Show', cut to three tokens. The bigram 'national western' is gold; this trigram was not listed. NLTK trigram chi-square score 5.83733e+06 at count 4 again rewards exclusive triples. It tracks trigram PMI's names, and it also admits code-like junk such as agenda numbers once their cells are extreme.

NLTK trigram t

#	phrase	score	f	label	commentary
1	be go to	9.4403	90	junk	Lemmatized 'is going to' or 'are going to'. The tagger turned the progressive into 'go'. Trigram t score 9.44029 at count 90 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.

#	phrase	score	f	label	commentary
2	i want to	8.5207	73	junk	The most common request preface in the transcripts. Trigram t score 8.52067 at count 73 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
3	thank you mr	8.2436	68	acceptable	The usual bridge into 'mr president'. Gold lists the bigram 'mr president', not this trigram. Trigram t score 8.24355 at count 68 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
4	i do not	7.4709	56	junk	Lemmatized negation, ordinary syntax. Trigram t score 7.47085 at count 56 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
5	be able to	7.0594	50	junk	Lemmatized 'be able to', an ordinary verb chain. Trigram t score 7.05936 at count 50 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
6	just want to	6.9229	48	junk	Request preface with 'just'. Trigram t score 6.92292 at count 48 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
7	that we be	6.6958	47	junk	Lemmatized 'that we are'. Trigram t score 6.69582 at count 47 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
8	one of the	6.5905	44	junk	A partitive frame. Trigram t score 6.59046 at count 44 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
9	i would like	6.5548	43	junk	A politeness preface, compositionally built. Trigram t score 6.55484 at count 43 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
10	would like to	6.5538	43	junk	Politeness chain, fully compositional. Trigram t score 6.55378 at count 43 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
11	and i be	6.5521	48	junk	Lemmatized 'and I am'. The copula was collapsed to 'be', so the trigram is an artifact. Trigram t score 6.55206 at count 48 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
12	thank you for	6.5400	43	acceptable	The formula plus the preposition that introduces the reason. Trigram t score 6.53997 at count 43 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
13	i just want	6.3204	40	junk	A discourse preface before a request. Trigram t score 6.32044 at count 40 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
14	i think that	6.3020	40	junk	Stance marker plus complementizer. Trigram t score 6.30204 at count 40 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
15	make sure that	5.9992	36	acceptable	The same discourse verb with a complementizer. Not gold. Trigram t score 5.99925 at count 36 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
16	that we have	5.8579	35	junk	A clause opening, not a term. Trigram t score 5.85789 at count 35 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.

#	phrase	score	f	label	commentary
17	and i think	5.7949	34	junk	A discourse opener, freely composed, not a taggable term. Trigram t score 5.79489 at count 34 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
18	go to be	5.7552	34	junk	Lemmatized 'going to be'. Trigram t score 5.75523 at count 34 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
19	you mr president	5.7440	33	acceptable	The address formula heard as 'thank you, Mr. President', without 'thank'. Gold kept 'mr president'. Trigram t score 5.74396 at count 33 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.
20	we be go	5.6224	32	junk	Lemmatized 'we are going'. Trigram t score 5.6224 at count 32 is the bigram t-test's frequency bias in a three-word window. Lemmatized verb chains dominate. The normal approximation is no better for trigram counts than for bigrams.

NLTK trigram G^2

#	phrase	score	f	label	commentary
1	thank you mr	4318.8138	68	acceptable	The usual bridge into 'mr president'. Gold lists the bigram 'mr president', not this trigram. Trigram G^2 score 4318.81 at count 68 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
2	to thank you	4002.3882	7	acceptable	The purpose clause in 'want to thank you'. Discourse, not a topic. Trigram G^2 score 4002.39 at count 7 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
3	thank you for	3966.5595	43	acceptable	The formula plus the preposition that introduces the reason. Trigram G^2 score 3966.56 at count 43 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
4	thank you councilman	3958.4386	19	acceptable	Politeness plus a title. Fixed as an address, not a topic. Trigram G^2 score 3958.44 at count 19 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
5	thank you madam	3940.2650	15	acceptable	Politeness plus a gendered form of address. Trigram G^2 score 3940.26 at count 15 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
6	thank you very	3908.5119	17	junk	'Thank you very much' cut before 'much'. The trigram is an incomplete intensifier. Trigram G^2 score 3908.51 at count 17 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
7	thank you counsel	3902.6079	12	acceptable	Politeness plus a mis-heard or alternate title 'counsel'. Trigram G^2 score 3902.61 at count 12 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
8	thank you counselor	3896.8226	7	acceptable	Politeness plus a title. Trigram G^2 score 3896.82 at count 7 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
9	thank you councilmember	3893.9436	14	acceptable	Politeness plus a title. Trigram G^2 score 3893.94 at count 14 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.

#	phrase	score	f	label	commentary
10	so thank you	3893.3944	24	acceptable	Politeness formula with a discourse 'so'. Trigram G^2 score 3893.39 at count 24 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
11	thank you councilor	3883.7837	9	acceptable	Politeness plus a title. Trigram G^2 score 3883.78 at count 9 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
12	thank you so	3882.4444	18	acceptable	Politeness with an intensifier starting the next bit. Trigram G^2 score 3882.44 at count 18 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
13	thank you and	3871.9067	5	junk	'Thank you' plus the coordinator that starts the next clause. The third word is not part of the formula. Trigram G^2 score 3871.91 at count 5 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
14	thank you ms	3871.6064	4	acceptable	Politeness plus a title. Trigram G^2 score 3871.61 at count 4 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
15	thank you president	3869.1780	7	acceptable	Politeness plus the office. Trigram G^2 score 3869.18 at count 7 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
16	thank you councilwoman	3865.8522	4	acceptable	Politeness plus a title. Trigram G^2 score 3865.85 at count 4 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
17	i thank you	3862.7548	4	acceptable	A politeness variant of 'thank you', not a domain tag. Trigram G^2 score 3862.75 at count 4 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
18	thank you to	3862.4652	8	junk	'Thank you' plus a preposition. The third token is whatever comes next, so the trigram is not fixed. Trigram G^2 score 3862.47 at count 8 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
19	say thank you	3861.2378	7	acceptable	A politeness frame, not a topic. Trigram G^2 score 3861.24 at count 7 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.
20	thank you again	3855.0766	3	acceptable	Politeness formula plus 'again'. Trigram G^2 score 3855.08 at count 3 collapses onto the 'thank you' family. Every extension of a very frequent bigram inherits a huge likelihood ratio. The list is non-directional dependence of one politeness formula, not a set of domain trigrams.

Failure analysis

Trigram extraction

Trigram PMI on the filtered set (frequency at least 3) reaches precision@20 of 0.8 against the 40 trigram gold phrases, with a mean top-20 frequency of 3.45. The head of that list is real institutional trigrams (`vince lombardi trophy`, `president pro tem`, `maximum building height`, `law enforcement oversight`, `mile high stadium`). It also lets in chopped committee titles. `civil right utility` and `right utility

economic` are not phrases anyone says. They are slices of the Seattle committee name "Civil Rights, Utilities, Economic Development" after lemmatization and after the comma was deleted. `national restaurant stock` is the same kind of damage around the National Western Stock Show. Trigram mining breaks down exactly where a transcript lists coordinated nouns: the window of three tokens is shorter than the name, so the scorer promotes the boundary pieces.

The unfiltered trigram PMI top-20 is entirely frequency 1 (fraction 1). With one attestation, position-specific marginals equal the joint count and trigram PMI saturates. Alphabetical tie-breaking then decides the list. That is not a ranking of domain phrases.

NLTK's trigram t-test and trigram likelihood ratio do not have this sparsity pattern, and they do not recover gold either. Trigram t precision@20 is 0. Its top row is `be go to`, the lemmatized form of "is going to" / "are going to". Trigram G^2 precision@20 is 0, and the whole top-20 is a paraphrase family of `thank you` plus the next word (`thank you mr`, `thank you madam`, `thank you councilmember`). G^2 is non-directional and scales with count, so the politeness formula crowds out every planning trigram. Three-word formulas that contain function words (`motion to approve`, `those in favor`, `in term of`) are real and are on the gold list, but they lose to either hapax exclusivity (PMI) or the thank-you family (G^2 and, for the t-test, the going-to family).

A phrase found by only one metric

Among the eight from-scratch bigram rankings, `council member` (frequency 69) is inside only the log-likelihood top-20. It is a genuine role title and it is on the gold list. PMI does not rank it in the top 20 because too many rarer exclusive pairs score higher. Student's t does not rank it because `thank you` and `of the` have larger t statistics. Chi-square's top 20 is already full of near-perfect rare pairs whose statistic sits near N. Likelihood ratio is the metric that still has room, at frequency 69, for a high-count title whose expected count under independence is much smaller than 69. `cape verdean` (frequency 12) is the Yates-only top-20 item. It is a real demonym tied to the Cabo Verde hearings, and it is not on the gold list. Yates's continuity correction reorders the near-tie at the top of Pearson's chi-square enough to admit it and to drop some frequency-3 names. `this be` is a t-test-only top-20 item: lemmatization collapsed "this is" to a pseudo-collocation, and the t statistic then rewards the high count.

Low counts, which dominate voice memos

On the unfiltered run, PMI's top-20 fraction at frequency 1 is 1, and the mean frequency of that top-20 is 1. Every one of those pairs is a hapax whose two words occur nowhere else, so PMI equals log2(N) and the alphabetical sort of the ties produces the list (`abstention iraqi`, `alma mater`, `astrobiologists democratization`, and so on). Some of those hapaxes are real names. Many are accidents of one sentence. The filtered run (frequency at least 3) is what keeps PMI usable: its filtered top-20 mean frequency is 3.9, precision@20 is 0.85, and the score-versus-frequency Spearman on the filtered set is -0.0705573. The negative sign is the low-frequency preference that remains even after the floor of 3.

Pearson's chi-square does the same thing when the filter is removed. A perfectly exclusive pair of any frequency, including 1, has chi-square equal to N (61728 here), so the unfiltered top is again an alphabetical hapax list (fraction at frequency 1: 1). That is the concrete sense in which chi-square does not penalize high-frequency noise the way a reader might hope, and also does not prefer high frequency at the very top: the statistic saturates. The textbook frequency effect shows up once pairs are not exclusive. `of the` is enormously common and only moderately dependent, so its chi-square is large in absolute terms but still below the saturated names. Yates changes the low-count story. Subtracting N/2 inside the absolute deviation demotes frequency-1 exclusive pairs. The unfiltered Yates top-20 has frequency-1 fraction 0 and is headed by the same mid-count names as the filtered list (`dry ice`, `boot camp`, `domestic violence`). The correction is doing the work the frequency filter does for Pearson.

Student's t does not need a hapax to go wrong. Filtered and unfiltered top-20s match, mean frequency 203, precision@20 0. Spearman of t against frequency on the filtered candidate set is 0.587808. The Manning and Schütze denominator sqrt(f(w1 w2)) is the Poisson variance-equals-mean assumption, and

the numerator grows with the raw count whenever the pair beats independence by a modest ratio. Common grammatical bigrams therefore outrank rare domain terms. The normality assumption behind a t statistic is not met by these counts: bigram indicators are sparse and skewed, which is why a rank order driven by t should not be read as a calibrated test. Log-likelihood is more tolerant of sparse counts than Pearson's chi-square in Dunning's argument, and on this corpus it is the from-scratch metric that surfaces `council member`, `long beach`, and `mr president` inside the top 20. It still spends most of that top 20 on `thank you`, `of the`, and `you know` (precision@20 0.15, hits 3) because G^2 is non-directional and still increases with sample size. The phrase list and the check that the t formula matches Manning and Schütze and NLTK are in the section below. The reason chi-square and G^2 can share a high rank correlation and still disagree in that top 20 is the section after it.

PMI^2 and PMI^3 do not rescue the low-count problem. On the positive head of the PMI list, x^2 and x^3 are strictly increasing, so the filtered top-20 of PMI, PMI^2, and PMI^3 is the same 20 phrases (precision@20 0.85 for PMI^2). PMI^3 preserves the entire ranking of PMI (Spearman 1). PMI^2 does not: a negative PMI becomes a positive score. The most negative filtered pair, `the to`, has PMI about -4.6 at frequency 5 and a PMI^2 rank in the hundreds, not the top 20, because 4.6 squared is still far below a PMI of 14 squared. The sign flip is real and it would pollute a cutoff set on PMI^2 if the threshold were pushed down the list. It does not change the top 20 of this corpus.

Student's t is a frequency-bias failure, and the formula is the documented one

The score in `src/models.py` is Manning and Schütze (1999) equation 5.10, t = (f(w1 w2) − f(w1) f(w2) / N) / sqrt(f(w1 w2)). N is the bigram-token count 61728, the same N the contingency table uses, not the word-token count 67046. On all 3864 filtered candidates, our t and NLTK `student_t` on the same cells have maximum absolute difference 0 and Spearman correlation 1. Direct substitution of the counts for `thank you` (f=488, f(w1)=524, f(w2)=1129, N=61728) returns 21.65687852. The same substitution for `of the` (f=396, f(w1)=1559, f(w2)=3155) returns 15.89554949. Both match our column and NLTK's column exactly.

The filtered top 20, with joint counts, is conversational filler and lemmatized copulas, not domain terms: `thank you` (488), `of the` (396), `want to` (210), `and i` (271), `i be` (306), `you know` (172), `in the` (256), `i think` (167), `we be` (234), `we have` (177), `that we` (180), `go to` (159), `do not` (140), `the city` (146), `so i` (136), `this be` (148), `there be` (114), `it be` (137), `i would` (93), `on the` (130). None is on the bigram gold list, so precision@20 is 0 (hits 0). The numerator grows with the raw excess count and the denominator only with the square root of the joint count, so these high-frequency pairs outrank rare names. `i be`, `we be`, `this be`, `there be`, and `it be` are that same bias after lemmatization collapsed "is" and "are" onto `be`.

Chi-square and G^2 are asymptotically equivalent and still disagree at the top

Pearson's chi-square and Dunning's G^2 share a chi-square null distribution with 1 degree of freedom on a 2x2 table with fixed margins. That is a statement about the null as N grows. It does not make the two scores monotone transforms of each other on the strong associations that occupy a collocation ranking. Over the 3864 filtered candidates the Spearman correlation between our chi-square and our G^2 is 0.964364. The two top-20 lists share one phrase, `long beach`. Chi-square precision@20 is 0.95 with mean top-20 frequency 8.4. G^2 precision@20 is 0.15 with mean top-20 frequency 166.5.

The implementations are the standard statistics. Rebuilding every filtered 2x2 table and comparing with scipy: our Pearson value versus `chi2_contingency(table, correction=False)` has median absolute difference 8.88178e-16 and maximum 7.27596e-12 (Spearman 0.9999999999; the gap from 1 is one tie moved by that floating-point residue, and no table differs by more than 1e-8). Our Yates value versus `correction=True` has median absolute difference 4.44089e-16 and maximum 1.45519e-11 (Spearman 1). Our G^2 versus `chi2_contingency(table, correction=False, lambda_="log-likelihood")` has median and maximum absolute difference 0 (Spearman 1). Scipy's default `correction=True` applies Yates before the G statistic, so that call is not Dunning's G^2; against it our G^2 has median absolute difference 2.01502 and maximum 15.7377. On the same cells, NLTK `chi_sq` matches our Pearson (maximum

absolute difference 7.27596e-12) and NLTK `likelihood_ratio` matches our G^2 (maximum 2.27374e-13). N in every table is the bigram count 61728. The four cells sum to that N. It is not the word-token count 67046.

The head splits for a concrete reason, and the vast O22 cell is not what inflates chi-square. An exclusive pair (O12 = O21 = 0, both marginals equal to the joint count) has Pearson chi-square equal to N at every frequency, because E11 = O11^2 / N is tiny and (O11 − E11)^2 / E11 saturates at N. Five phrases from the chi-square top 20, written as f(w1 w2), f(w1), f(w2), chi-square, G^2:

- `boot camp`: 4, 4, 4, 61728, 85.153329. Chi-square rank 1, G^2 rank 236. Of the Pearson sum, the O11 term is 61720.0003 and the O22 term is 0.0003. Of G^2, the O11 term 2·O11·ln(O11/E11) is 77.1536 (90.6%) and the O22 term is 7.9997. E11 is 0.000259. - `coffee shop`: 3, 3, 3, 61728, 65.591138. G^2 rank 333. Same exclusive ceiling. - `dry ice`: 8, 8, 9, 54868.4443, 152.936807. One stray count keeps it just under N. - `mm hmm`: 6, 6, 7, 52908.8571, 117.122590. - `san francisco`: 6, 7, 6, 52908.8571, 117.122590.

Eighty-five percent of that top 20 has a structural zero in O12 or O21. The list is low-count near-exclusive names (median frequency 5), not high-frequency grammar.

G^2 = 2 · Σ Oij ln(Oij / Eij) has no ceiling at N. Its largest piece on these tables is the O11 term, which scales with the joint count times the log lift, so frequent conversational pairs outrank exclusive names. Five phrases from the G^2 top 20, same columns:

- `thank you`: 488, 524, 1129, 24535.0907, 3890.292701. G^2 rank 1, chi-square rank 25 (0.397·N). E11 is 9.5839 and O11/E11 is 50.919. The O11 term is 3835.90 (98.6% of G^2). The O22 term is 960.63 and is partly cancelled by negative O12 and O21 terms. Pearson's O22 term is 3.81 (0.016% of chi-square). - `want to`: 210, 228, 2311, 4958.1658, 1274.005384. Chi-square rank 143, G^2 rank 2. - `you know`: 172, 996, 312, 5657.0308, 1025.240110. Chi-square rank 128, G^2 rank 3. - `i think`: 167, 1625, 221, 4602.7152, 988.987220. Chi-square rank 154, G^2 rank 4. - `do not`: 140, 527, 438, 5043.7588, 829.281463. Chi-square rank 142, G^2 rank 5.

So the 0.95 versus 0.15 precision@20 gap is a property of the two statistics on this corpus. Chi-square's top is whatever sits nearest the ceiling N, which here is medium-to-low frequency near-exclusive pairs. G^2's top is whatever maximizes the O11 log-lift term, which here is high-frequency filler. The bulk ranking still agrees (Spearman 0.964364). Nothing in the marginals, in O22, or in the log base was wrong.

Gold-list incompleteness

The bigram gold list has 104 phrases and the trigram list has 40. Both were checked for frequency at least 3. Neither is exhaustive. Precision@20 divides by 20, so any true collocation that was not labeled counts as a miss and pulls precision down. That makes every precision@20 in the comparison table a lower bound, not an estimate of error against all true municipal collocations.

The bound is visible in the lists. `mm hmm` is a fair miss: it is a backchannel. `core mission`, `santa cruz`, `tara rise`, `cape verdean`, and `annual action plan` are fixed expressions that were not labeled. Counting them as junk would overstate the error of PMI, NPMI, and chi-square. `thank you` and `you know` are fixed discourse formulas and were deliberately left off the domain list; calling them junk would also be wrong, and calling them gold would have handed free hits to the t-test and to G^2. The qualitative labels keep those cases separate from grammatical junk (`of the`, `in the`) and from lemmatization junk (`this be`, `i be`, `be go to`).

Stopword removal is not free precision. Only 2 bigram gold phrases contain a stopword (`motion to`, `move to`), but the ablation changes the whole contingency table. After stopwords are deleted, Student's t precision@20 rises from 0 to 0.55 (11 hits) because `of the` can no longer occupy the top. PMI's precision stays high and does not need that deletion. The gain for t is real, and the cost is every formula that used a function word, including `in term of`.

What this evaluation shows, and what it does not

It shows, on 67046 processed tokens of real council speech, that the eight scores do not rank the same phrases. PMI, NPMI, and chi-square concentrate on exclusive compounds and names. Their filtered precision@20 values are 0.850, 0.950, and 0.950. Student's t precision@20 is 0.000 with mean top-20 frequency 203.0. Log-likelihood precision@20 is 0.150 and is the only from-scratch list that contains council member. It shows that NLTK's formulas match ours when the cells match, and that finder.N does not match our N. It shows that a frequency floor of 3 changes PMI and Pearson chi-square from an alphabetical hapax list into a list of repeated names, and that Yates does part of that job without the floor.

It does not show that NPMI or chi-square is a calibrated tagger. Nineteen of twenty top phrases landing on the gold list is partly a property of this gold list, which includes rare exclusive names and excludes thank you. A gold list built only from high-frequency institutional titles would have reversed the precision ranking. It does not show significance. No p-value cutoff was applied, and the t-test's normal model is already known to be the wrong sampling story for these counts. It does not show trigram extraction working as a general phrase chunker: windows of three tokens slice committee titles, and the unfiltered trigram PMI list is entirely frequency 1. It does not transfer, without a new gold list, to a different genre of voice memo. The license is non-commercial. The sample is one split and 28 segments.